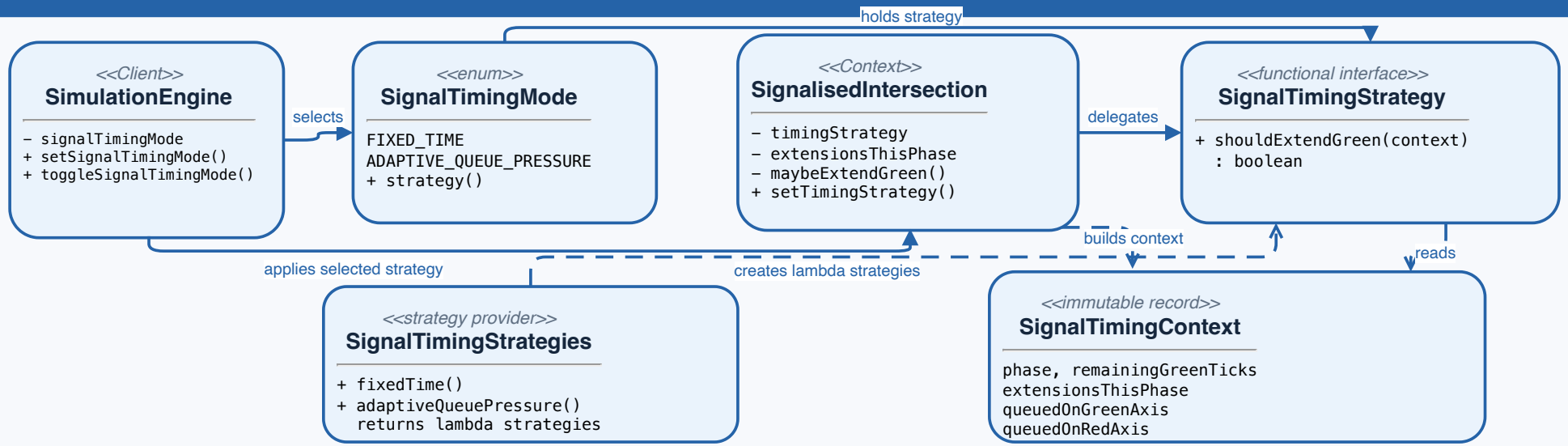
COMP2000 Assignment 2 - Design Pattern Implementation

Traffic simulation I Key participants and runtime interactions I Editable UML view

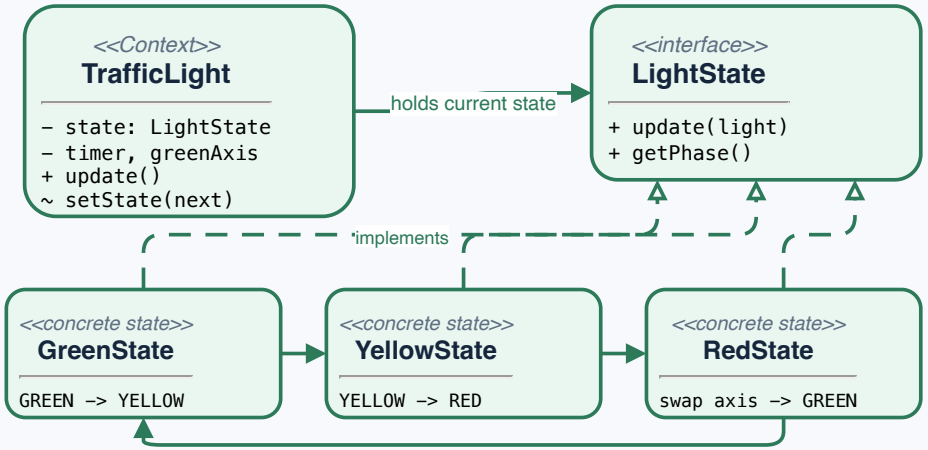
1. STRATEGY - Switchable traffic-signal timing (Assignment 2 focus)



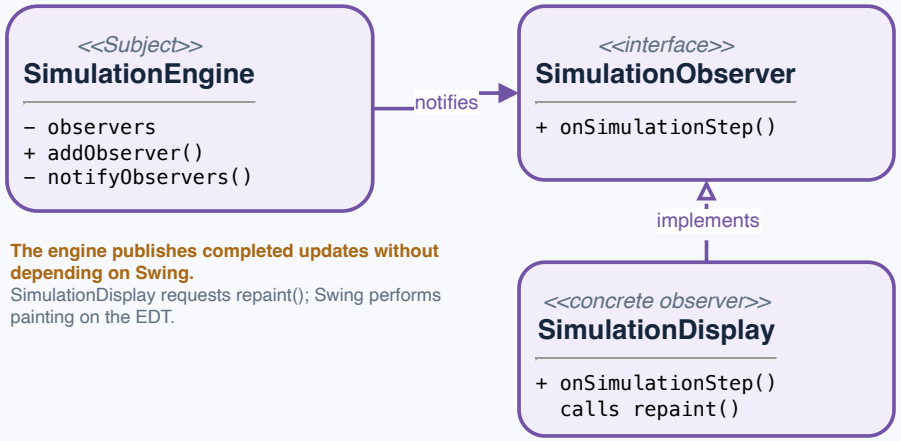
Runtime flow: engine selects mode -> intersection builds context -> strategy decides extension.

2. STATE - Traffic-light phase lifecycle

TrafficLight delegates update(); each state owns its transition rule.



3. OBSERVER - Simulation-to-display updates



The engine publishes completed updates without depending on Swing.
SimulationDisplay requests repaint(); Swing performs painting on the EDT.

Solid arrow: runtime association / call I Dashed arrow: dependency / realization

Based on the final assignment-2-development implementation